

Muhammad Hassan

AI Engineer | Agentic and Multimodal Systems

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Technical Skills

Languages: Python, C/C++, TypeScript, JavaScript, HTML/CSS

Agents and Automation: LangChain, LangGraph, LangSmith, OpenRouter, AI APIs & SDK

Product Engineering: Next.js, React, Zustand, Tailwind CSS, Node.js, Express.js, Socket.IO, Web RTC, Cypress

Databases: Pinecone, Weaviate, PostgreSQL, MongoDB, Firebase, Supabase, CMS (Strapi, Sanity & Payload), Prisma

DevOps: Redis, Kafka, Docker, AWS (EC2, S3, Amplify & Lambda), Kubernetes, Jenkins

Experience

AI Engineer | *Useryze*

Dec 2024 – Present

- Built an end-to-end **AI content system** with RAG using OpenAI, LangChain, Pinecone, Next.js, and Prisma. Delivered the authenticated source-to-draft pipeline following ingestion, document splitting, vector embeddings, semantic retrieval, generation, editing, persistence, and export using scraped content from social media channels by python scraping jobs.
- Developed the organization's landing page as CRO for A/B testing following its deployment on VPS & AWS Amplify. Powered all website through Payload CMS for dynamic content throughout.

Full-Stack Engineer | *GT Solutions USA*

May 2024 – Jun 2025

- Built full-stack and mobile application systems for complex operational workflows, including role-based CRM products, for Daycares in USA, serving 1,000+ staff.
- Created an Admin Panel for **Aga Khan Hospital** as part of their research for a food survey project. Added necessary functionalities (e.g. location access, offline data management etc) into the mobile app.

Junior Full-Stack Developer | *Xerverse.io*

Jun 2023 – Nov 2023

- Delivered the responsive frontend for **DataPlus**, an AI-data platform providing human-verified, rights-secured, and training-ready datasets; built reusable content sections, interactive animations, and conversion flows that communicate data quality, compliance, and model-readiness.

Projects

Restaurant Voice and Ordering Agent

Personal

OpenAI, Vapi, WebRTC, Socket.IO, Next.js, Node.js

- Built a real-time **voice ordering agent** that maintains conversation context and invokes structured tools to inspect menus, validate items, create and modify orders, calculate bills, and track order status.
- Integrated Vapi/WebRTC with an event-driven browser call flow for live transcripts, microphone controls, call-state recovery, and responsive speech feedback.

Khaata360 – AI Personal Finance Assistant

Academic

OpenRouter, FastAPI, Next.js, Twilio, MongoDB, Docker

- Developed a bilingual **AI finance agent** for WhatsApp and web that classifies intent, extracts transaction entities, manages multi-turn state, and answers natural-language queries over user financial data.
- Added vision-based **receipt extraction**, AI-generated financial charts, analytics dashboards, verified phone linking, and containerized FastAPI/Next.js services with MongoDB persistence.

Awards

Full-Stack Hackathon Winner – Zabefest '24

May 2024

SZABIST

- Built a secure cashier-less checkout app in 48 hours with QR/manual product entry, location verification, authentication, and protected routes.

Web Development Runner-Up – Developer's Day '24

Apr 2024

FAST-NUCES

- Created a GitHub-like version-control platform supporting repositories, organizations, commits, history tracking, and admin rollback.

Education

Habib University

BS Computer Science